



# Analysis of HIV-1 BF *pr/rt* recombinant strains from Rio de Janeiro/Brazil reveals multiple unrelated mosaic structures

Monick Lindenmeyer Guimarães\*, José Carlos Couto-Fernandez, Walter de Araujo Eyer-Silva, Sylvia Lopes Maia Teixeira, Saada Lima Chequer-Fernandez, Mariza Gonçalves Morgado

Laboratory of AIDS and Molecular Immunology, Oswaldo Cruz Institute – FIOCRUZ, Rio de Janeiro, Brazil

## ARTICLE INFO

### Article history:

Received 5 March 2010

Received in revised form 24 June 2010

Accepted 2 July 2010

Available online 16 July 2010

### Keywords:

Brazil

HIV-1

URF<sub>BF1</sub>

CRF<sub>BF1</sub>

Recombination

## ABSTRACT

Due to the co-circulation of HIV-1 subtypes B and F1 in Brazil, a large variety of unique BF1 recombinant forms (URFs<sub>BF1</sub>) and four circulating BF1 recombinant forms (CRF28, CRF29, CRF39 and CRF40) have been described. The aim of this study was to investigate mosaic structure and phylogenetic relationship among several BF1 (protease–reverse transcriptase, *pr/rt*) recombinant sequences obtained from a group of patients from Rio de Janeiro, Brazil, from 1994 to 2005. Phylogenetic relationships were estimated by Bayesian and SplitsTree methods. Recombination breakpoints were analyzed by Bootscan. Those samples presenting the same recombinant pattern in the *pr/rt* region were investigated in the integrase (*int*) and envelope (*env*) regions as a screening method for the detection of potential new CRFs candidates. Third out of 61 *pr/rt* HIV-1 BF1 recombinant sequences analyzed depicted unique recombinant structures and were classified URFs<sub>BF1</sub>. The other 31 samples segregated in eight well-supported phylogenetic clusters composed of at least three samples sharing the same recombination pattern. Analyses of the *int* and *env* regions from these 31 samples revealed that 11 samples were URFs<sub>BF1</sub>. Three and four sequences corresponded, respectively, to the previously described CRF39 and CRF40. Three samples displayed a CRF28-like mosaic structure, and one sample a CRF29-like mosaic pattern. The other nine BF1 samples segregate in three distinct clusters with the same recombination profile and could represent good candidates for new CRFs<sub>BF</sub> profiles. The HIV-1 BF1 epidemic in Rio de Janeiro is characterized by a high prevalence (67%) of URFs<sub>BF1</sub> and a low prevalence (4.9–6.6%) of each CRFs<sub>BF</sub> previously identified in Brazil.

© 2010 Elsevier B.V. All rights reserved.

## 1. Introduction

It has been shown that at least 10% of the HIV-1 circulating strains comprise inter-subtype recombinants (McCutchan, 2000), and this event has an important impact on HIV-1 evolution. However, the number of recombinant strains could be previously underestimated, as initial epidemiological studies focused just on the envelope region, where the occurrence of recombination phenomenon is rare. More recently, with the advent of HIV antiretroviral therapy monitoring, an ever-increasing number of protease (*pr*) and reverse transcriptase (*rt*) sequences have become available and adopted for phylogenetic inferences. This has significantly increased the description of inter-subtype recombination events, as important recombination hotspots have been

identified in this region, as well as in the integrase and *gp41* regions, suggesting a direct correlation between recombination frequency and sequence similarity throughout the HIV-1 genome (Magiorkinis et al., 2003; Thomson and Najera, 2005).

Areas where more than one HIV-1 subtype co-circulate in high frequency are referred as “geographic recombination hotspots” (Thomson et al., 2004), with detection of a great number of unique recombinant forms (URFs) and/or circulating recombinant forms (CRFs). Brazil has been considered to be one of the “geographic recombination hotspots”, showing an overall predominance of HIV-1 subtypes B, F1 and BF1 recombinants in most country regions. In Rio de Janeiro subtype B samples account for ~80% of the *pol* analyzed sequences, followed by subtype F1 (2.7–5.7%) and BF1 recombinants and a very small proportion of other subtypes (Brindeiro et al., 2003; Sá-Ferreira et al., 2007).

The introduction of subtypes B and F1 in the country was estimated between 1965–1970 and 1975–1980, respectively (Bello et al., 2007; Bello et al., 2006). These subtypes seem to have experienced an initial period of exponential growth, on average doubling every one or two years (Bello et al., 2007). Thus, as a consequence of the co-circulation and rapid expansion of these

\* Corresponding author at: Laboratory of AIDS and Molecular Immunology, Oswaldo Cruz Institute, FIOCRUZ, Av. Brasil 4365, Leonidas Deane Building, Room #413, Rio de Janeiro 21045-900, Brazil. Tel.: +55 21 3865 8106; fax: +55 21 3865 8173.

E-mail address: [monicklg@ioc.fiocruz.br](mailto:monicklg@ioc.fiocruz.br) (M.L. Guimarães).

subtypes in the country, the occurrence of a myriad of URFs\_BF1 stand was documented as a major feature of the Brazilian epidemic (Brennan et al., 2007; Brindeiro et al., 2003; Cabral et al., 2006; Cavalcanti et al., 2007; De Sá Filho et al., 2006; Guimarães et al., 2002; Sá-Ferreira et al., 2007; Sa Filho et al., 2005; Stefani et al., 2007; Veras et al., 2007; Vicente et al., 2000). In addition, four CRFs\_BF1 (CRF28, CRF29, CRF39 and CRF40) were described (De Sá Filho et al., 2006; Guimarães et al., 2008).

The presence of the HIV-1 BF1 recombinant virus was first described in Brazil in the early 1990s (Morgado et al., 1994; Sabino et al., 1994). Although other studies had documented the presence of BF1 *pol* recombinants in Rio de Janeiro (Brindeiro et al., 2003; Sá-Ferreira et al., 2007), little is known about their mosaic structures, the relationship among them and with the CRFs\_BF1 already described, or their epidemiological relevance.

The present study aimed to characterize the mosaic structure of samples identified as BF1 in the *pr/rt* region collected in the Rio de Janeiro State, Southeastern region of Brazil, from 1994 to 2005, and also to investigate the phylogenetic relationship among these samples and the CRFs\_BF1 previously described.

## 2. Materials and methods

### 2.1. Study population

Sixty-one HIV-1 BF1 recombinant strains were identified after screening of a total of 1504 *pr/rt* sequences recovered from 1994 to 2005 in HIV-1 seropositive individuals from Rio de Janeiro State. All samples were classified in the *pr/rt* region in previous studies carried out in our laboratory (Couto-Fernandez et al., 2005; Eyer-Silva et al., 2007a,b; Maia Teixeira et al., 2006). The BF1 samples were mostly obtained (82%) from patients under highly active antiretroviral therapy (HAART) collected from 2002 to 2005. The remaining ones (18%) were collected from 1994 to 2005. The risk group was not available for the most patients. Sequences were excluded if they came from individuals known to be related by a direct transmission or from the same patient. Blood collection was performed after informed consent as specified in each of the original studies (Couto-Fernandez et al., 2005; Eyer-Silva et al., 2007a,b; Maia Teixeira et al., 2006). The re-analyses of the *pr/rt* sequences and DNA sequencing of the *int* and *env* regions from the BF1 selected samples followed the Brazilian legislation for Ethics in Research with Human Beings.

### 2.2. HIV-1 *pr/rt* sequences

Genomic DNA was extracted from whole blood of drug naïve patients using a DNA extraction column kit (QIAamp DNA Mini Blood Kit; QIAGEN, Valencia, CA) and sequencing using a home brew protocol method as described elsewhere (Eyer-Silva et al., 2007a,b; Maia Teixeira et al., 2006). HIV-1 plasma RNA genotyping of *pr/rt* region of patients under highly active antiretroviral therapy were obtained using the ViroSeq HIV-1 Genotyping System protocol (Celera Diagnostic, Abbott Laboratories, Des Plaines, IL) (Couto-Fernandez et al., 2005). Phylogenetic trees of *pr/rt* sequences from patients under HAART were previously constructed by the maximum likelihood method with PAUP, version 4.0b10 (Swofford, 2003). For the other patients phylogenetic inferences were performed by Neighbor-Joining (NJ) algorithm as previously described (Eyer-Silva et al., 2007a,b; Maia Teixeira et al., 2006). Inferences of recombination breakpoints were performed by bootscan analysis (Salminen et al., 1995), using the SimPlot software.

### 2.3. HIV-1 integrase and envelope sequences

In order to better analyze the complexity of the BF1 recombinant patterns circulating in Rio de Janeiro, the samples

belonging to the clusters detected in the *pr/rt* region were selected for further investigation based on the *env-gp120* (*env-1*), *int* and *env-gp41* (*env-2*) as they are considered as highly recombinogenic regions (Magiorkinis et al., 2003; Thomson and Najera, 2005). Amplification of these regions was performed as described elsewhere (Passaes et al., 2009). All PCR products were purified using Illustra GFX PCR DNA purification kit (GE Healthcare, Amersham, Buckinghamshire, UK) according to the manufacturer's protocol. Purified DNA was sequenced by using the ABI BigDye Terminator version 3.1 cycle sequencing ready reaction kit (Applied Biosystem, CA, U.S.A.) and processed with an automated ABI 3100 Genetic Analyzer (Applied Biosystem, CA, U.S.A.).

### 2.4. Sequence alignments

Nucleotide sequences of the studied strains were aligned with subtypes B, F1 and CRFs\_BF1 previously described using the Clustal X program (Thompson et al., 1997) and later manually edited for minor adjustments. All positions with alignment gaps were removed. Four distinct alignments were used in the present study: the *pr/rt* alignment covered a fragment of 1231 nt (2277–3507 positions relative to HXB2 strain), the second alignment of around 864 nt covered the complete *int* (nucleotides 4230–5093 relative to HXB2); the third alignment of 404 nt spans the *env-1* region (nucleotides 6876–7282 relative to HXB2), and the fourth alignment of 593 nt spans the *env-2* region (nucleotides 7620–8209 relative to HXB2). All alignments are available from the authors upon request.

### 2.5. Sequence analysis

The phylogenetic relationship within the BF1 studied samples and between them and the CRFs\_BF1 described were assessed by Bayesian method and confirmed SplitsTree analysis. The jModeltest 0.1.1 program was used to select the best-fit model of nucleotide substitution under the Akaike information criteria, resulting in the choice of the GTR+ I + G model (Tavaré, 1986) as implemented in MrBayes version 3.1.2 (Ronquist and Huelsenbeck, 2003). For each data set, two runs of 4 chains each (one cold and three heated, temp = 0.20) were run for  $1 \times 10^7$  generations, with a burn-in of  $1 \times 10^6$  generations. Convergence of parameters was assessed by calculating the effective sample size (ESS) using TRACER version 1.4 (<http://beast.bio.ed.ac.uk/Tracer/>) excluding an initial 10% for each run. All parameter estimates for each run showed an ESS values >100. A final Bayesian majority-rule consensus tree was obtained for each data set.

The SplitsTree program version 4 (Huson and Bryant, 2006) was employed to confirm the phylogenetic relationship of BF1 Brazilian samples in the *pr/rt* region and the method used was NeighborNet based on the pairwise distance estimated by F84 parameter model.

Bootscan analysis (Salminen et al., 1995) was used to detect recombination as implemented in the SimPlot software version 3.5.1 (<http://www.med.jhu.edu/deptmed/sray/download/>). The analyses were performed on a sliding window of 200 nt moving in steps of 20 nt using the K2-parameter model, with representative HIV-1 strains representatives of group M subtypes. To better characterize the recombination breakpoints suggested in the previous analyses, the putative recombinants were subjected to informative site analyses (Sierra et al., 2005) and were finally confirmed by separate Neighbor-Joining trees constructed using the K2-parameter model in the MEGA 4.0 program (Tamura et al., 2007).

### 2.6. Sequence data

GenBank accession numbers: HM439694–HM439716 (*env-2*), HM439716–HM439738 (*env-1*), HM439739–HM439762 (*int*).

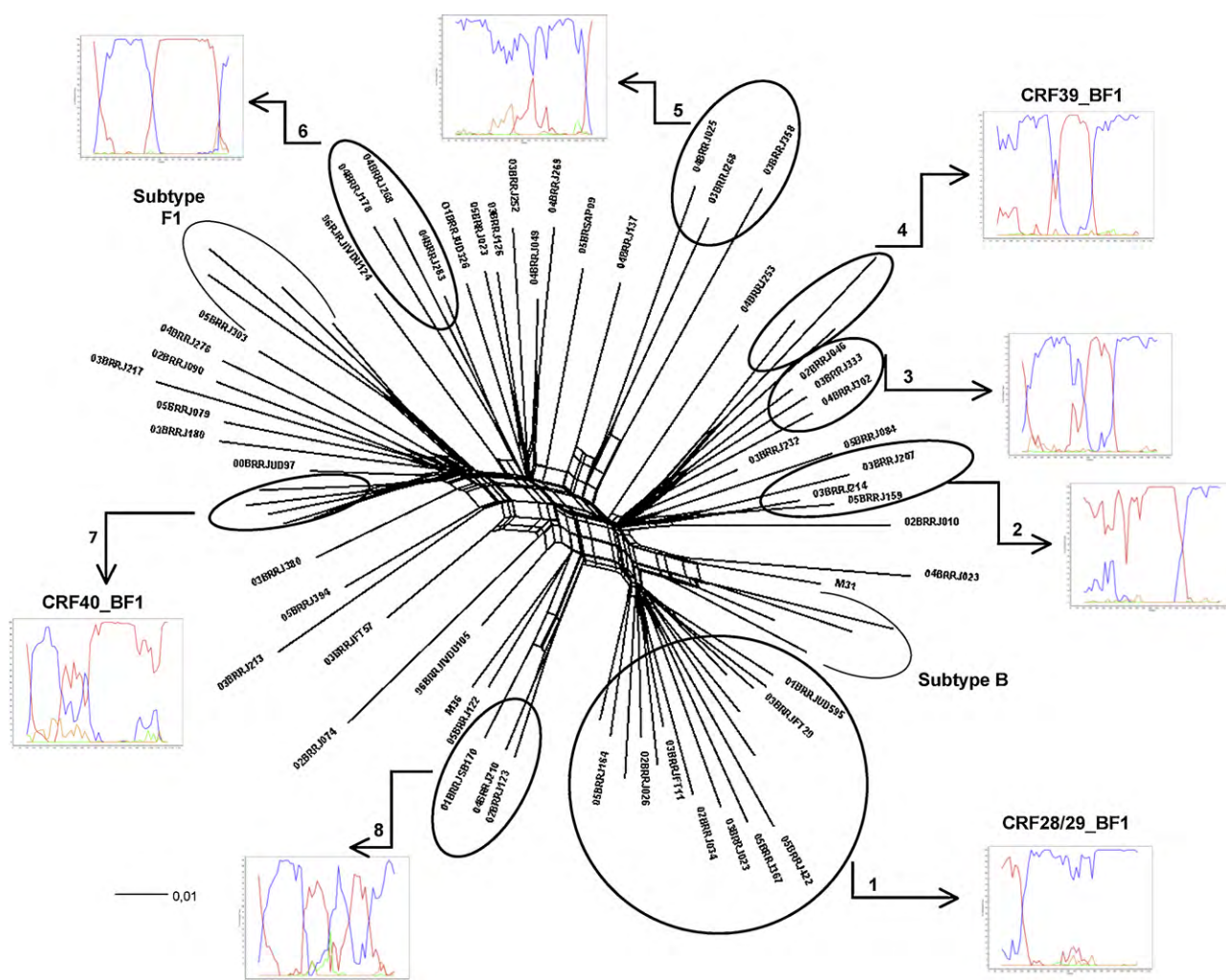
### 3. Results

From a total number of 1504 HIV-1 *pr/rt* nucleotide sequences obtained from 1994 to 2005, in Rio de Janeiro State, Brazil, 61 (4.1%) presented a BF1 recombinant pattern. In order to better characterize the mosaic structure and the relationships among these BF1 *pr/rt* recombinant samples and those CRF\_BF already described a combination of Bayesian, SplitsTree, bootscan and informative site analysis was carried out. According to these analyses, a complex diversity of BF1 recombination patterns was verified and a total of 30 samples displaying a unique recombination profile were classified as URFs\_BF. The remaining 31 sequences segregates in eight well-supported clusters composed of at least three samples sharing the same recombination patterns (Fig. 1). The criteria used for the definition of these clusters were: (a) clusters with posterior probability values higher than 9 in the Bayesian analysis, (b) clusters converging to the same summit in the SplitsTree and (c) same mosaic pattern in the bootscan analysis.

Samples from cluster 1 formed a monophyletic group with the CRF28\_BF and CRF29\_BF, as both CRFs present the same

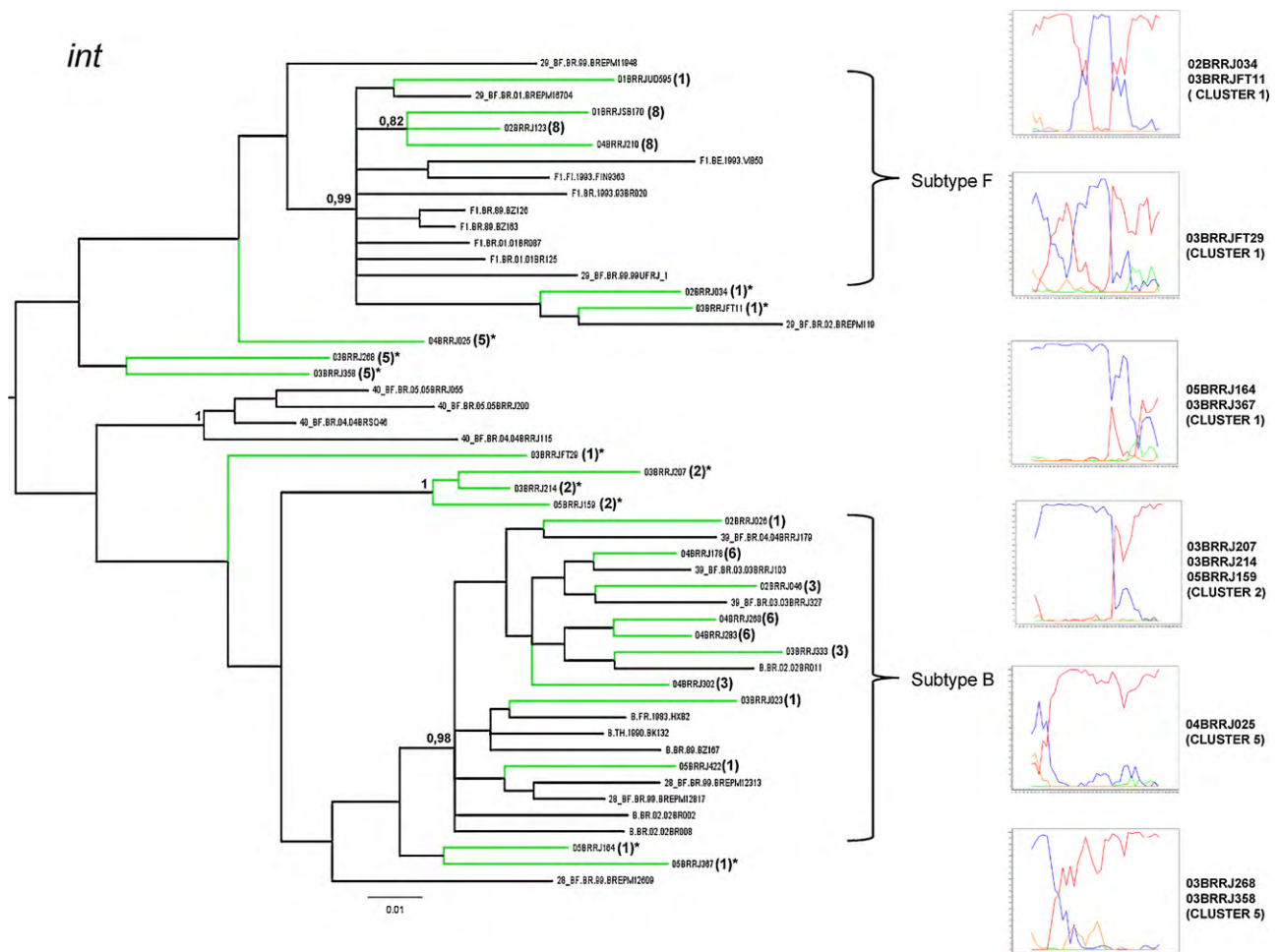
recombination pattern in this genomic region (Fig. 1). Samples from clusters 4 and 7 correspond, respectively, to the newly described CRF39\_BF and CRF40\_BF and were fully characterized elsewhere (Guimarães et al., 2008). Samples from cluster 3, segregate in a branch related to the CRF39 and possess the same recombination pattern of this CRF except for the inclusion of one F fragment in the beginning of the analyzed region (2277–2474, relative to HXB2) (Fig. 1), suggesting that these sequences could share a common BF1 recombinant ancestor with the CRF39\_BF. The other four clusters displayed distinct BF1 patterns, as could be visualized in the bootscan analysis in the Fig. 1. None of the studied samples grouped with the CRFs\_BF described in other South American countries (CRF12, CRF17, and CRF38) (data not shown).

In order to better analyze the complexity and the phylogenetic relationships among BF1 recombinant forms circulating in Rio de Janeiro, those 24 HIV-1 samples belonging to *pr/rt* clusters 1–3, 5, 6 and 8 were selected for additional analysis of the *int* (Fig. 2) and the *env* regions (Fig. 3). Samples of cluster 1 exhibited diverse genomic structures in the *int* region: three samples (02BRRJ026, 03BRRJ023, and 05BRRJ422) were subtype B, one sample (01BRRJUD595) was



**Fig. 1.** SplitsTree analysis of the HIV-1 Brazilian BF1 *pol* sequences (1231 nt) recovered from patients' residents in Rio de Janeiro, Brazil, from 1994 to 2005. The splits graph was constructed using NeighborNet methodology with pairwise distance input, which was estimated by F84 distance. Alignment was prepared with HIV-1 group M subtypes B and F1, as well as Brazilian CRF\_BF1 forms (Los Alamos HIV sequence database). Eight clusters with at least three sequences that were subsequently found to share the same inter-subtype breakpoints are indicated. The scale bar indicates 10% nucleotide sequence divergence. Free branches indicate reference sequences. Bootscanning plots depicting the relationship of BF1 inter-subtype recombinant clusters (1–8) are represented. Bootscan analysis was performed on a sliding window of 200 nt of the query sequences moving along an alignment of reference sequences by increments of 20 nt. Reference samples used for these analyses were: subtype B (BZ163, blue), subtype F1 (BZ163, red) subtype C (92BR025, orange), and subtype A1 (U455, green).

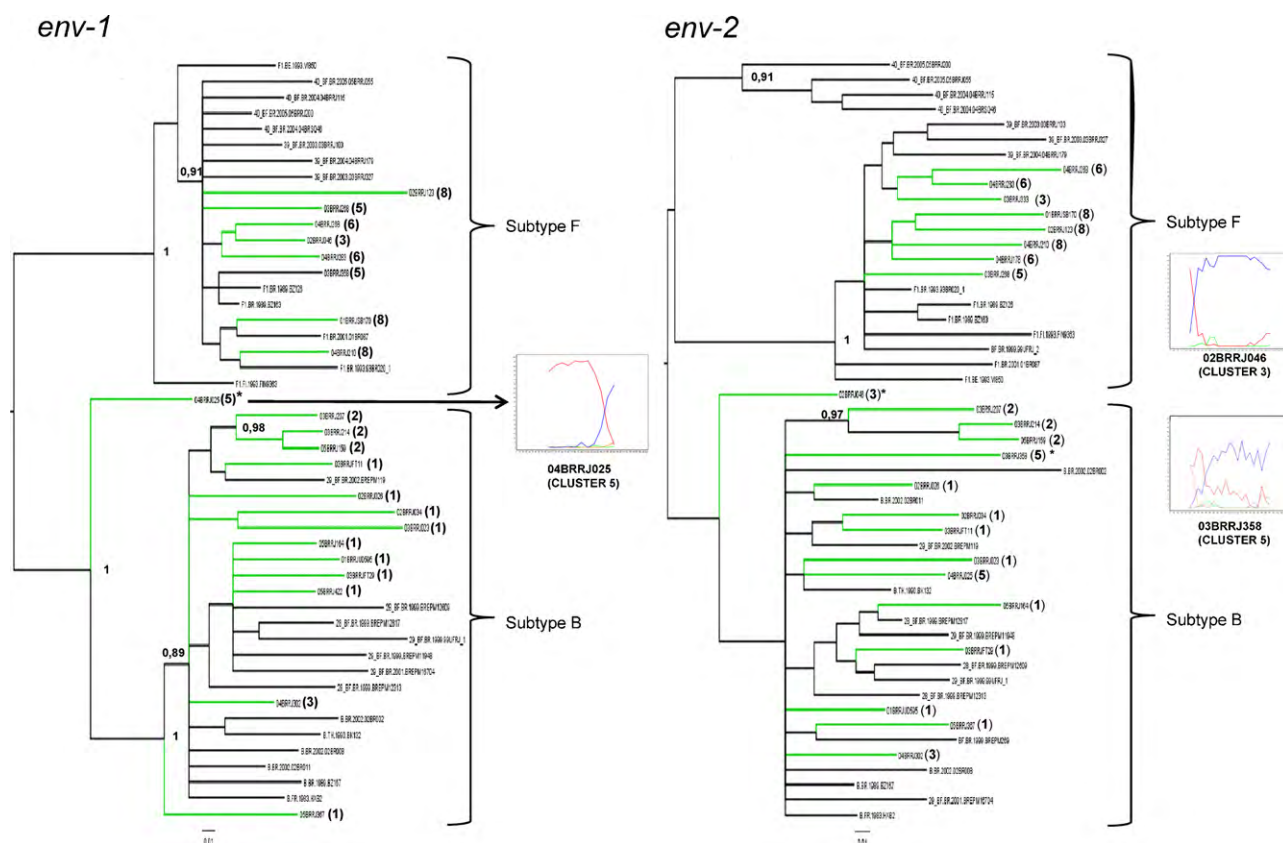




subtype F1, and five samples (02BRRJ034, 03BRRJFT11, 05BRRJ164, 05BRRJ367, and 03BRRJFT29) were classified as BF1 recombinants, with three different recombination patterns (Fig. 2). All samples from cluster 1 were classified as subtype B in the *env-1* and *env-2* regions (except for 05BRRJ422 that did not amplify) (Fig. 3). Samples from cluster 2 displayed the same BF1 recombination pattern in the *int* region (Fig. 2) and were classified as subtype B in the *env-1* and *env-2* regions (Fig. 3). Samples from cluster 3 were classified as subtype B in the *int* region (Fig. 2) but present divergent subtype assignments in the *env-1* and *env-2* regions (Fig. 3) as follows: sample 04BRRJ302 was classified as subtype B in both regions; sample 02BRRJ046 was classified as subtype F1 and BF1 in the *env-1* and *env-2* regions, respectively; whereas sample 03BRRJ333 did not amplify in *env-1* and was subtyped as F1 in *env-2*. Samples from cluster 5 were classified as BF1 recombinants in the *int* region with two samples (03BRRJ268 and 03BRRJ358) displaying the same recombination profile and the other one (04BRRJ025) displaying a shorter subtype B fragment (Fig. 2). Samples from this cluster also displayed divergent subtype assignments in the *env-1* and *env-2* regions (Fig. 3): sample 03BRRJ268 was classified as subtype F1 in both regions; sample 03BRRJ358 was classified as subtype F1 and BF1 in the *env-1* and *env-2* regions, respectively; whereas sample 04BRRJ025 was classified as BF1 and B in the *env-1* and *env-2*

## 4. Discussion

After an extensive screening of 1504 HIV-1 samples from Rio de Janeiro State, Southeastern Brazil, we were able to identify 61 (4.1%) samples as BF1 recombinants in the *pr/rf* region. Similar proportions of BF1 recombinants were detected in independent studies conducted in Rio de Janeiro (4%) (Sá-Ferreira et al., 2007), São Paulo (4.1%) (Gonsalez et al., 2007), Federal District (Central Brazil) (3.1%) (Veras et al., 2007) and Northeast region (4%) (Cavalcanti et al., 2007). Other studies had documented higher proportion of BF1 recombinants in Minas Gerais (6%) (Waleria-Aleixo et al., 2008), São Paulo (7.3–8.4%) (Barreto et al., 2006; Sa Filho et al., 2005), Santos (28.9%) (de Sa-Filho et al., 2008) and Goiania (7.2–15%) (Cardoso et al., 2009; Cardoso and Stefani, 2009).



**Fig. 3.** Majority-rule Bayesian consensus tree of the *env-1* (404 nt) and *env-2* (593nt). Posterior probability values are indicated only at key nodes. Samples were numbered according to the *pr/rt* classification. Samples from the present work have their branch highlighted in green. (\*) indicates intragenic BF1 recombinant samples in the *env-1* or *env-2* region. Bootscan analyses were performed as described in Fig. 2.

showing that the frequency of recombinants vary across different country regions. Higher proportions (~15%) of BF1 recombinants have been also observed in intravenous drug user samples collected in Rio de Janeiro from 1994 to 2001 (Maia Teixeira et al., 2006), indicating that the frequency of these recombinants could differ according to the HIV transmission route.

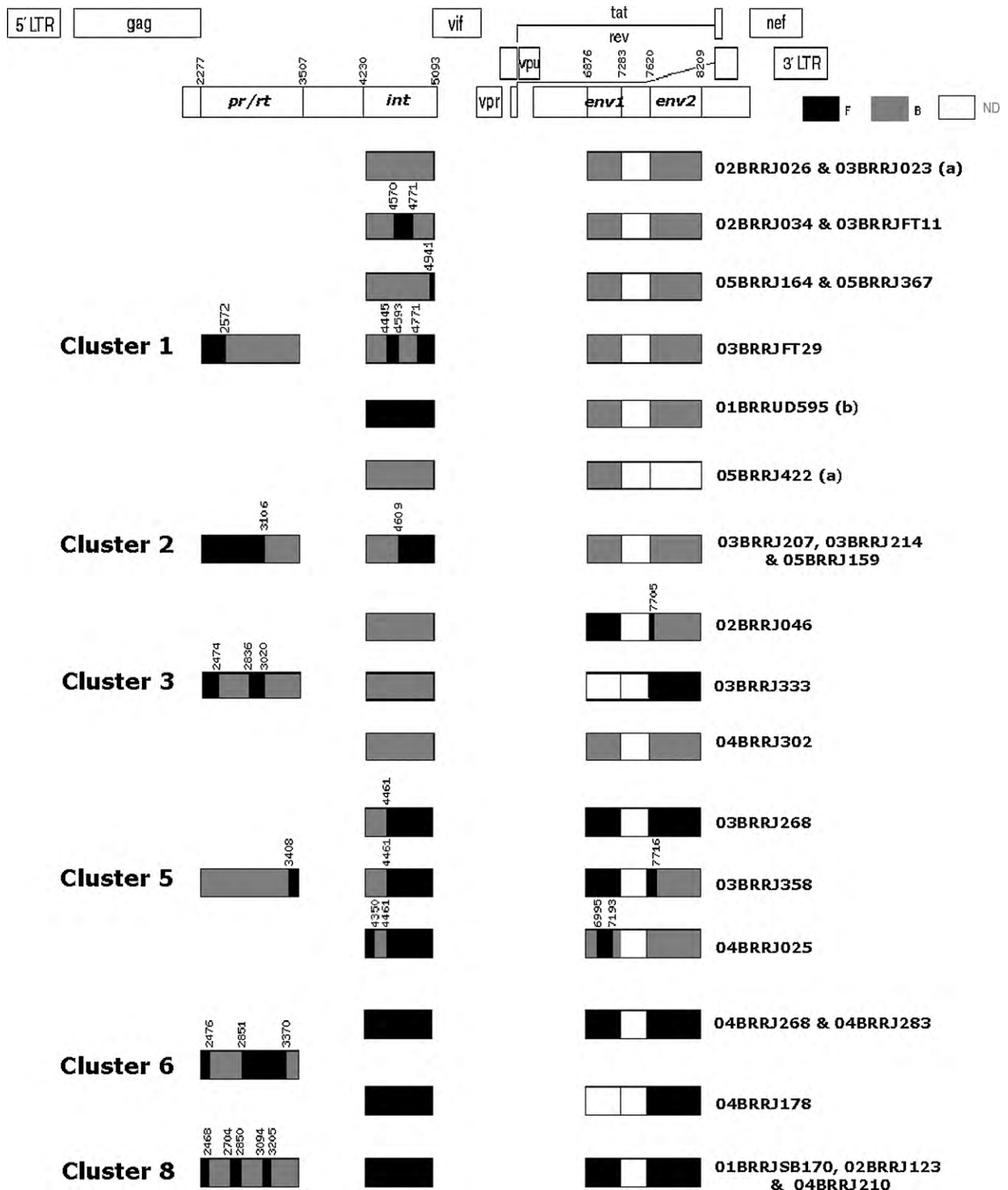
According to the genetic analysis of the 61 BF1 samples from Rio de Janeiro, 41 sequences corresponded to URFs<sub>BF</sub> (67%) and 20 could be divided according to the recombination patterns as CRFs<sub>BF</sub>s or potential candidates to CRFs<sub>BF</sub> (33%). From those, three samples (4.9%) displayed a CRF28<sub>BF</sub>-like mosaic pattern and one sample (1.7%) a CRF29<sub>BF</sub>-like mosaic pattern in the analyzed regions. Three (4.9%) and four (6.6%) samples were classified as CRF39<sub>BF</sub> and CRF40<sub>BF</sub>, respectively (Guimarães et al., 2008). Finally, nine samples segregate in three distinct phylogenetic groups (clusters 2, 5 and 6). Each cluster contains three sequences with the same mosaic structures in the four genomic regions analyzed and could represent new CRFs. It is important to highlight that, if these three potential new CRFs do not finally share the same structure when entirely sequenced, the URF<sub>BF</sub> prevalence would increase from 67% to 82% among BF1 recombinant samples.

The lower epidemiological relevance of each CRF<sub>BF</sub> in Rio de Janeiro contrasts with the epidemiological profile in two other cities from the Southeastern region of Brazil, Santos (de Sa-Filho et al., 2008) and São Paulo (de Souza et al., 2008), where the CRF28<sub>BF</sub> and CRF29<sub>BF</sub> *pol* structures were found to represent 83% and 30% of the BF1 recombinant samples, respectively. The extremely high prevalence of those CRFs<sub>BF</sub> in Santos might be a consequence of the importance of intravenous drug users in the local epidemic, which contrasts with the epidemiological profile of HIV infection in Rio de

Janeiro city where sexual transmission is the predominant route. CRFs<sub>BF</sub> also plays an important role in the local epidemics of other South American countries like as CRF12<sub>BF</sub> in Argentina (Quarleri et al., 2004) and the CRF12<sub>BF</sub> and CRF38<sub>BF</sub> in Uruguay (Ruchansky et al., 2009). A report concerning samples from Chile revealed that BF1 recombinants in *pol* represent 15% and most of them grouped into two clusters, one related to the CRF12<sub>BF</sub> and the other possibly representing a new CRF (Rios et al., 2007).

Even including those nine possible CRFs<sub>BF</sub> candidates (samples from clusters 2, 5, and 6) as URFs<sub>BF</sub>, this group would account for only 3.3% of the analyzed samples, similar to those verified in São Paulo (3%) and Santos (4.8%) (de Souza et al., 2008; de Sa-Filho et al., 2008). Taken together, the low prevalence of both URFs<sub>BF</sub> and CRFs<sub>BF</sub> in Rio de Janeiro has a direct correlation to the lower subtype F1 (5%) circulation, leading to a minor probability of inter-subtype recombination events. This findings contrasts with Argentinian molecular epidemiology, where 56.9% samples from newly diagnosed individuals (2003–2005) where BF1 recombinants (22.7% of URFs<sub>BF</sub>, 15.55% of CRF12<sub>BF</sub> and 15.55% CRF12<sub>BF</sub> related) (Dilernia et al., 2007). In Argentina subtype F is rare, being mostly present in their recombinant form BF1. Thus, we suggest that the high prevalence of both CRFs<sub>BF</sub> and URFs<sub>BF</sub> in these geographic regions might results from founder events, such as from a CRF12<sub>BF</sub> or an ancestor related form.

In summary, our study reveals that most of the BF1 recombinants result of independent recombination events between subtypes B and F1 and that these URFs<sub>BF</sub> are an important component of the HIV-1 epidemic in Rio de Janeiro State, Brazil. The prevalence of each CRFs<sub>BF</sub> individually seems to be very low in this state, which contrast with the epidemiological profile of



**Fig. 4.** Schematic representation of mosaic structures of six clusters of samples depicted in *pr/rt* analyses. Breakpoints patterns based on the analyses of the *pr/rt* (2277–3507 nt relative to HXB2), *int* (4230–5093 nt relative to HXB2), *env-1* (6876–7282 nt relative to HXB2) and *env-2* (7620–8209 nt relative to HXB2) regions. Subtype F fragments are represented in black, B fragments in gray and those fragments not determined (ND) are in white. (a) Indicate sequences with the same recombination profile as CRF28\_BF and (b) CRF29\_BF in all fragments analyzed in this study.

some neighboring cities like Santos (de Sa-Filho et al., 2008) and São Paulo (de Souza et al., 2008). These results support the notion that the HIV-1 epidemic in Brazil is heterogeneous and varies not only across distinct country regions but also among cities within the same geographic region.

#### Acknowledgments

We acknowledge Dr Gonzalo Bello and Caroline P.B. Passaes for helpful discussions and for critical reviewing of the manuscript. We acknowledge Dr Vera Bongertz for review and suggestions.



This study was supported in part by FIOCRUZ and Brazilian Ministry of Health and CNPq/PROSUL.

## Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.meegid.2010.07.001.

## References

- Barreto, C.C., Nishiyama, A., Araujo, L.V., Ferreira, J.E., Busch, M.P., Sabino, E.C., 2006. Trends in antiretroviral drug resistance and clade distributions among HIV-1 infected blood donors in Sao Paulo, Brazil. *J. Acquir. Immune Defic. Syndr.* 41, 338–341.
- Bello, G., Eyer-Silva, W.A., Couto-Fernandez, J.C., Guimaraes, M.L., Chequer-Fernandez, S.L., Teixeira, S.L., Morgado, M.G., 2007. Demographic history of HIV-1 subtypes B and F in Brazil. *Infect. Genet. Evol.* 7, 263–270.
- Bello, G., Guimaraes, M.L., Morgado, M.G., 2006. Evolutionary history of HIV-1 subtype B and F infections in Brazil. *AIDS* 20, 763–768.
- Brennan, C.A., Brites, C., Bodelle, P., Golden, A., Hackett Jr., J., Holzmayer, V., Swanson, P., Vallari, A., Yamaguchi, J., Devare, S., Pedrosa, C., Ramos, A., Badaro, R., 2007. HIV-1 strains identified in Brazilian blood donors: significant prevalence of B/F1 recombinants. *AIDS Res. Hum. Retroviruses* 23, 1434–1441.
- Brindeiro, R.M., Diaz, R.S., Sabino, E.C., Morgado, M.G., Pires, I.L., Brigido, L., Dantas, M.C., Barreira, D., Teixeira, P.R., Tanuri, A., 2003. Brazilian network for HIV drug resistance surveillance (HIV-BResNet): a survey of chronically infected individuals. *AIDS* 17, 1063–1069.
- Cabral, V.P., Cunha, C.B., Magalhaes, E.F., Pinto-Neto, L.F., Couto-Fernandez, J.C., Dietze, R., Morgado, M.G., Ribeiro-Rodrigues, R., 2006. Human immunodeficiency virus type-1 subtypes of infected patients in Espirito Santo, Brazil. *Mem. Inst. Oswaldo Cruz* 101, 881–885.
- Cardoso, L.P., Queiroz, B.B., Stefani, M.M., 2009. HIV-1 pol phylogenetic diversity and antiretroviral resistance mutations in treatment naive patients from Central West Brazil. *J. Clin. Virol.* 46, 134–139.
- Cardoso, L.P., Stefani, M.M., 2009. High level of multidrug resistance mutations in HIV type 1 pol gene and resistance-associated mutations to enfuvirtide (T-20) among antiretroviral-experienced patients from central Brazil. *AIDS Res. Hum. Retroviruses* 25, 943–950.
- Cavalcanti, A.M., Lacerda, H.R., Brito, A.M., Pereira, S., Medeiros, D., Oliveira, S., 2007. Antiretroviral resistance in individuals presenting therapeutic failure and subtypes of the human immunodeficiency virus type 1 in the Northeast Region of Brazil. *Mem. Inst. Oswaldo Cruz* 102, 785–792.
- Couto-Fernandez, J.C., Silva-de-Jesus, C., Veloso, V.G., Rachid, M., Gracie, R.S., Chequer-Fernandez, S.L., Oliveira, S.M., Arakaki-Sanchez, D., Chequer, P.J., Morgado, M.G., 2005. Human immunodeficiency virus type 1 (HIV-1) genotyping in Rio de Janeiro, Brazil: assessing subtype and drug-resistance associated mutations in HIV-1 infected individuals failing highly active antiretroviral therapy. *Mem. Inst. Oswaldo Cruz* 100, 73–78.
- de Sa-Filho, D.J., Soares Mda, S., Candido, V., Gagliani, L.H., Cavaliere, E., Diaz, R.S., Caseiro, M.M., 2008. HIV type 1 pol gene diversity and antiretroviral drug resistance mutations in Santos, Brazil. *AIDS Res. Hum. Retroviruses* 24, 347–353.
- De Sá Filho, D.J., Sucupira, M.C., Caseiro, M.M., Sabino, E.C., Diaz, R.S., Janini, L.M., 2006. Identification of two HIV type 1 circulating recombinant forms in Brazil. *AIDS Res. Hum. Retroviruses* 22, 1–13.
- de Souza, A.C., de Oliveira, C.M., Rodrigues, C.L., Silva, S.A., Levi, J.E., 2008. Short communication: molecular characterization of HIV type 1 BF pol recombinants from Sao Paulo, Brazil. *AIDS Res. Hum. Retroviruses* 24, 1521–1525.
- Dilernia, D.A., Gomez, A.M., Lourttau, L., Marone, R., Losso, M.H., Salomón, H., Gómez-Carrillo, M., 2007. HIV type 1 genetic diversity surveillance among newly diagnosed individuals from 2003 to 2005 in Buenos Aires, Argentina. *AIDS Res. Hum. Retroviruses* 23, 1201–1207.
- Eyer-Silva, W.A., Bello, G., Morgado, M.G., 2007a. Molecular evidence that human immunodeficiency virus type 1 dissemination in a small Brazilian city was already taking place in the early 1990s. *Mem. Inst. Oswaldo Cruz* 102, 647–649.
- Eyer-Silva, W.A., Couto-Fernandez, J.C., Morgado, M.G., 2007b. Molecular epidemiology of HIV type 1 in inner Rio De Janeiro State, Brazil. *AIDS Res. Hum. Retroviruses* 23, 303–308.
- Gonzalez, C.R., Alcalde, R., Nishiyama, A., Barreto, C.C., Silva, F.E., de Almeida, A., Mendonça, M., Ferreira, F., Fernandes, S.S., Casseb, J., Duarte, A.J., 2007. Drug resistance among chronic HIV-1-infected patients naive for use of anti-retroviral therapy in Sao Paulo city. *Virus Res.* 129, 87–90.
- Guimarães, M.L., dos Santos Moreira, A., Loureiro, R., Galvao-Castro, B., Morgado, M.G., 2002. High frequency of recombinant genomes in HIV type 1 samples from Brazilian southeastern and southern regions. *AIDS Res. Hum. Retroviruses* 18, 1261–1269.
- Guimarães, M.L., Eyer-Silva, W.A., Couto-Fernandez, J.C., Morgado, M.G., 2008. Identification of two new CRF\_BF in Rio de Janeiro State, Brazil. *AIDS* 22, 433–435.
- Huson, D.H., Bryant, D., 2006. Application of phylogenetic networks in evolutionary studies. *Mol. Biol. Evol.* 23, 254–267.
- Magiorkinis, G., Paraskevis, D., Vandamme, A.M., Magiorkinis, E., Sypsa, V., Hatzakis, A., 2003. In vivo characteristics of human immunodeficiency virus type 1 intersubtype recombination: determination of hot spots and correlation with sequence similarity. *J. Gen. Virol.* 84, 2715–2722.
- Maia Teixeira, S.L., Bastos, F.I., Hacker, M.A., Guimaraes, M.L., Morgado, M.G., 2006. Trends in drug resistance mutations in antiretroviral-naïve intravenous drug users of Rio de Janeiro. *J. Med. Virol.* 78, 764–769.
- McCutchan, F.E., 2000. Understanding the genetic diversity of HIV-1. *AIDS* 14 (Suppl. 3), S31–44.
- Morgado, M.G., Sabino, E.C., Shpaer, E.G., Bongertz, V., Brigido, L., Guimaraes, M.D., Castilho, E.A., Galvao-Castro, B., Mullins, J.I., Hendry, R.M., et al., 1994. V3 region polymorphisms in HIV-1 from Brazil: prevalence of subtype B strains divergent from North American/European prototype and detection of subtype F. *AIDS Res. Hum. Retroviruses* 10, 569–576.
- Passaes, C.P., Bello, G., Lorete, R.S., Matos Almeida, S.E., Junqueira, D.M., Veloso, V.G., Morgado, M.G., Guimaraes, M.L., 2009. Genetic characterization of HIV-1 BC recombinants and evolutionary history of the CRF31\_BC in Southern Brazil. *Infect. Genet. Evol.* 9, 474–482.
- Quarleri, J.F., Rubio, A., Carobene, M., Turk, G., Vignoles, M., Harrigan, R.P., Montaner, J.S., Salomon, H., Gomez-Carrillo, M., 2004. HIV type 1 BF recombinant strains exhibit different pol gene mosaic patterns: descriptive analysis from 284 patients under treatment failure. *AIDS Res. Hum. Retroviruses* 20, 1100–1107.
- Rios, M., Delgado, E., Perez-Alvarez, L., Fernandez, J., Galvez, P., de Parga, E.V., Yung, V., Thomson, M.M., Najera, R., 2007. Antiretroviral drug resistance and phylogenetic diversity of HIV-1 in Chile. *J. Med. Virol.* 79, 647–656.
- Ronquist, F., Huelsenbeck, J.P., 2003. MrBayes 3: Bayesian phylogenetic inference under mixed models. *Bioinformatics* 19, 1572–1574.
- Ruchansky, D., Casado, C., Russi, J.C., Arbiza, J.R., Lopez-Galindez, C., 2009. Identification of a new HIV Type 1 circulating recombinant form (CRF38\_BF1) in Uruguay. *AIDS Res. Hum. Retroviruses* 25, 351–356.
- Sá-Ferreira, J.A., Brindeiro, P.A., Chequer-Fernandez, S., Tanuri, A., Morgado, M.G., 2007. Human immunodeficiency virus-1 subtypes and antiretroviral drug resistance profiles among drug-naïve Brazilian blood donors. *Transfusion* 47, 97–102.
- Sa Filho, D.J., Sanabani, S., Diaz, R.S., Munerato, P., Brunstein, A., Fusuma, E., Sabino, E.C., Janini, L.M., 2005. Analysis of full-length human immunodeficiency virus type 1 genome reveals a variable spectrum of subtypes B and f recombinants in Sao Paulo, Brazil. *AIDS Res. Hum. Retroviruses* 21, 145–151.
- Sabino, E.C., Shpaer, E.G., Morgado, M.G., Korber, B.T., Diaz, R.S., Bongertz, V., Cavalcante, S., Galvao-Castro, B., Mullins, J.I., Mayer, A., 1994. Identification of human immunodeficiency virus type 1 envelope genes recombinant between subtypes B and F in two epidemiologically linked individuals from Brazil. *J. Virol.* 68, 6340–6346.
- Salminen, M.O., Carr, J.K., Burke, D.S., McCutchan, F.E., 1995. Identification of breakpoints in intergenotypic recombinants of HIV type 1 by bootscanning. *AIDS Res. Hum. Retroviruses* 11, 1423–1425.
- Sierra, M., Thomson, M.M., Rios, M., Casado, G., Castro, R.O., Delgado, E., Echevarria, G., Munoz, M., Colomina, J., Carmona, R., Vega, Y., Parga, E.V., Medrano, L., Perez-Alvarez, L., Contreras, G., Najera, R., 2005. The analysis of near full-length genome sequences of human immunodeficiency virus type 1 BF intersubtype recombinant viruses from Chile, Venezuela and Spain reveals their relationship to diverse lineages of recombinant viruses related to CRF12\_BF. *Infect. Genet. Evol.* 5, 209–217.
- Stefani, M.M., Pereira, G.A.S., Lins, J.A.B., Alcantara, K.C., Silveira, A.A., Viegas, A.A., Maya, N.C., Mussi, A.H., 2007. Molecular screening shows extensive HIV-1 genetic diversity in Central West Brazil. *J. Clin. Virol.* 39, 205–209.
- Swofford, D.L., 2003. PAUP: phylogenetic analysis using parsimony (M and other methods), version 4.0b 10. Sinauer Associates, Sunderland, MA.
- Tamura, K., Dudley, J., Nei, M., Kumar, S., 2007. MEGA4: molecular evolutionary genetics analysis (MEGA) software version 4.0. *Mol. Biol. Evol.* 24, 1596–1599.
- Tavaré, S., 1986. Some probabilistic and statistical problems in the analysis of DNA sequences. In: Miura, R.M. (Ed.), *Some Mathematical Questions in Biology—DNA Sequence Analysis*. American Mathematical Society, Providence (RI), pp. 57–86.
- Thompson, J.D., Gibson, T.J., Plewniak, F., Jeanmougin, F., Higgins, D.G., 1997. The CLUSTAL\_X windows interface: flexible strategies for multiple sequence alignment aided by quality analysis tools. *Nucleic Acids Res.* 25, 4876–4882.
- Thomson, M.M., Najera, R., 2005. Molecular epidemiology of HIV-1 variants in the global AIDS pandemic: an update. *AIDS Rev.* 7, 210–224.
- Thomson, M.M., Sierra, M., Tanuri, A., May, S., Casado, G., Manjon, N., Najera, R., 2004. Analysis of near full-length genome sequences of HIV type 1 BF intersubtype recombinant viruses from Brazil reveals their independent origins and their lack of relationship to CRF12\_BF. *AIDS Res. Hum. Retroviruses* 20, 1126–1133.
- Veras, N.M., Veras, V.S., Ramalho, E.D., Kyaw, C., Silva, R.R., Brigido, M.M., Martins, C.R., 2007. HIV type 1 genetic variability in central Brazil. *AIDS Res. Hum. Retroviruses* 23, 1481–1490.
- Vicente, A.C., Otsuki, K., Silva, N.B., Castilho, M.C., Barros, F.S., Pieniazek, D., Hu, D., Rayfield, M.A., Bretas, G., Tanuri, A., 2000. The HIV epidemic in the Amazon Basin is driven by prototypic and recombinant HIV-1 subtypes B and F. *J. Acquir. Immune Defic. Syndr.* 23, 327–331.
- Waleria-Aleixo, A., Martins, A.N., Arruda, M.B., Brindeiro, R.M., Da-Silva, R.M., Nobre, F.F., Greco, D.B., Tanuri, A., 2008. Drug resistance mutation profile and accumulation kinetics in human immunodeficiency virus-positive individuals infected with subtypes B and F failing highly active antiretroviral therapy are influenced by different viral codon usage patterns. *Antimicrob. Agents Chemother.* 52, 4497–4502.